

Print Estimating and Order Management Software – Project Overview

Primary Goal:

To create a flexible, clean, and efficient estimating and job management system that supports multiple printing sectors (wide-format, digital, folded carton/packaging, flexo) and is customizable for different company workflows. It should be compatible with both cloud-based web app and standalone Mac applications.

Core Features

1. Multi-Company Support

- **Company Profiles:**
 - Each company has unique settings, price lists, customers, and jobs.
 - Admin panel to manage company accounts and users.
 - Option to switch between companies easily.

2. Customer Management

- Add, edit, and view customer records (contact details, shipping/billing info).
- Track estimate and job history with filtering options.

3. Estimating

- Create custom estimates for different printing types (wide-format, digital, folded carton/packaging, flexographic).
- Input job specifications like size, quantity, materials, labor, and finishing.
- Apply pricing rules and save multiple versions of estimates.
- Convert approved estimates to active jobs with one click.

4. Order Management

- Create and track jobs with status workflow:
 - Estimate → Approved → In Production → Complete → Shipped.
- Attach artwork or proofs, assign internal notes, and track production and shipping steps.

5. Packing Slips & CoC

- Auto-generate packing slips from job data.
- Generate customizable Certificates of Conformance (CoC) based on job specifications.

6. Inventory Management

- Track stock materials (substrates, ink, packaging).
- Auto-deduct from jobs and set low-stock alerts.

- Track vendors and costs; manually upload or edit inventory.

7. Outside Vendor/Broker Section

- Mark jobs as outsourced and track vendor status and deliveries.
- Include outsourced costs in job pricing and profit tracking.

8. Customization & Flexibility

- Allow companies to:
 - Create custom job types, materials, processes, and pricing logic.
 - Adjust workflows, templates, and settings without requiring coding knowledge.

9. User Roles

- **Admin** – Full control over system settings and user management.
 - **Estimator** – Access to create and manage estimates.
 - **Production** – Access to job tickets and progress updates.
 - **Shipping** – Access to packing slips and job completion.
 - **Sales/View Only** – Limited access for sales staff.
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Deployment Options

1. Cloud-Based Web App

- Accessible via browsers from anywhere.
- Multi-user with secure login.
- Hosted on scalable infrastructure (AWS, Azure, etc.).

2. Standalone Mac App

- Runs locally on macOS with options for networked setups (local file sharing or internal LAN).
 - Sync option for future cloud-based version.
 - Compatible with macOS-native UI principles for smooth experience.
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Tech Stack & Requirements

Web App

- **Frontend:** React.js or Vue.js
- **Backend:** Node.js / Laravel / Django
- **Database:** PostgreSQL or MySQL (cloud-based)

Mac App

- **Tech:** Swift / SwiftUI (preferred) or Electron (acceptable for cross-platform compatibility)
- **Local Database:** SQLite

Cloud Syncing & Local Storage

- **Sync Mechanism:** Local SQLite database with API-based syncing to the cloud.
- **Security:** OAuth2 and role-based access control for user management.
- **PDF Generation:** Native macOS PDFKit for estimates, packing slips, and CoCs.

UI/UX

- **Design Principles:** Simple, fast UI with a desktop-first approach.
- **MacOS Feel:** Clean, minimal, professional design with smooth navigation, subtle shadows, and responsive interface. Color palette includes:
 - **Background:** #F4F4F4
 - **Panels/Cards:** #FFFFFF
 - **Text:** #333333
 - **Borders:** #DCDCDC
 - **Accent:** #4A90E2 (soft blue), secondary hover accents: #6E6E6E

Workflow Outline

1. **Login & Company Selection:**
 - Users select a company profile (e.g., "Premier Printing" or "SignFX").
 - Each company has distinct settings and workflows based on the print type (Flexo, Digital Commercial, Wide-format, etc.).
2. **Dashboard:**
 - Options to access:
 - Customers
 - Prospects
 - Estimating
 - Sales Reps
 - Jobs
 - Inventory
 - Vendors
3. **Customer Profiles:**
 - View, modify, and duplicate past orders.
 - Customizable fields based on the print industry type.
4. **Estimating & Orders:**
 - Industry-specific job workflows:
 - **Flexo:** Includes die libraries, imposition tools, ink usage.
 - **Digital Commercial:** Simpler estimating flow.
 - **Wide-Format:** Custom fields for materials and finishing.
 - Finalize estimates into jobs and track them through the production lifecycle.
5. **Order Tracking:**
 - Status tracking: Estimate → Approved → Production → Complete → Shipped.

- Sync job status with packing slips, certificates, and invoices.

Deliverables

- Fully functional cloud app + standalone Mac app.
- Admin tools for company and user management.
- Clean, maintainable source code and installer/build files.
- User guide or walkthrough.
- Future support and maintenance options.

With this outline, the designer can start working on the UI/UX, keeping in mind the flexibility, user roles, and company-specific customization. The development team can also use this structure to build both the web and standalone applications, ensuring smooth integration between the two deployment models.

Textual Wireframe for Print Estimating & Order Management System

This wireframe outlines the structure and navigation flow for the system. Each section represents a screen, its key components, and navigation links.

1. Login Screen

Components:

- Logo & App Name
- Company Selection Dropdown
- Username & Password Fields
- "Remember Me" Checkbox
- "Forgot Password?" Link
- "Login" Button

Navigation:

- After login → **Company Selection Screen**
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2. Company Selection Screen

(Only visible if the user is part of multiple companies)

Components:

- Header: "Select a Company"
- List of companies (e.g., "Premier Printing," "SignFX")
- "Proceed" Button

Navigation:

- Select a company → **Dashboard**
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3. Dashboard

Components:

- Sidebar Navigation:
 - Home
 - Customers
 - Prospects
 - Estimating
 - Orders
 - Inventory
 - Vendors
 - Reports
 - Settings (Admin only)
- Main Content Area:
 - Quick Stats (Pending Estimates, Open Jobs, Inventory Alerts)
 - Recent Activity Log
 - Shortcut Buttons (New Estimate, New Job, Add Customer)

Navigation:

- Click on any module → Respective Module Screen
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4. Customers Screen

Components:

- Search & Filter Customers
- List of Customers (Name, Contact, Last Order)
- Add New Customer Button
- Clickable Customer Profile

Customer Profile View:

- Customer Info (Editable)
- Order & Estimate History
- Notes & Attachments
- Duplicate Previous Order Button

Navigation:

- Click on a customer → **Customer Profile**
 - Click on "New Estimate" → **Estimate Creation Screen**
 - Click on "New Order" → **Order Creation Screen**
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5. Estimating Screen

Components:

- List of Estimates (Sortable by Date, Status, Amount)
- New Estimate Button

New Estimate Workflow:

1. Select Industry Type:
 - Flexo
 - Digital Commercial
 - Wide-Format
2. Enter Job Specifications (Size, Quantity, Materials, Finishing, etc.)
3. Cost Breakdown & Profit Margin Calculation
4. Generate PDF & Send to Customer
5. Convert to Job (if approved)

Navigation:

- Click "Generate PDF" → **PDF Preview Screen**
 - Click "Convert to Job" → **Job Tracking Screen**
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6. Orders Screen

Components:

- Search & Filter Orders
- Job Status Columns (Pending, In Production, Completed, Shipped)
- Clickable Job Details

Order Details View:

- Order Summary (Customer, Job Type, Pricing)
- Production Notes & Attachments
- Status Update (Move Job Through Stages)

Navigation:

- Click on an order → **Order Details**
 - Click "Update Status" → Order Status Updated
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7. Inventory Management

Components:

- List of Materials (Substrates, Inks, Packaging)
- Search & Filter Inventory
- Low Stock Alerts
- Add New Material Button
- Edit Costs & Vendor Assignments

Navigation:

- Click on a material → **Material Details Screen**
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8. Vendors Screen

Components:

- Vendor List (Name, Materials Supplied, Last Order Date)
- Add New Vendor Button
- Clickable Vendor Profile

Navigation:

- Click on a vendor → **Vendor Profile Screen**
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9. Reports Screen

Components:

- Generate Reports by Date/Type
 - Sales, Cost Analysis, and Profitability Reports
 - Export as CSV/PDF
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10. Settings (Admin Only)

Components:

- User Management (Add/Edit Users, Assign Roles)
 - Pricing Rules & Templates
 - System Customization (Themes, Branding)
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Navigation Flow Summary

1. **Login** → Company Selection → Dashboard
 2. **Dashboard** → Quick Access to Customers, Estimates, Orders, etc.
 3. **Customers** → Customer Profile → Order History → New Estimate/Job
 4. **Estimating** → Create Estimate → Generate PDF/Send → Convert to Job
 5. **Orders** → View Job Status → Update → Mark as Shipped
 6. **Inventory & Vendors** → Manage Materials & Suppliers
 7. **Reports & Settings** → System Customization & User Roles
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